

ANALISIS MEKANISME SISTEM REKOMENDASI JURUSAN

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1. ARSITEKTUR HYBRID

$$\text{HybridScore}(i) = (0.6 \times \text{Score_cbf}(i)) + (0.4 \times \text{Score_cf}(i))$$

2. CONTENT-BASED FILTERING (CBF)

Representasi Vektor:

$$S = (s_1, s_2, s_3, \dots, s_n)$$

$$J = (j_1, j_2, j_3, \dots, j_n)$$

Cosine Similarity:

$$\text{Similarity}(S, J) = (S \cdot J) / (\|S\| \times \|J\|)$$

Dot Product:

$$S \cdot J = \sum (s_k \times j_k)$$

Magnitude:

$$\|S\| = \sqrt{\sum (s_k^2)}$$

$$\|J\| = \sqrt{\sum (j_k^2)}$$

Rumus Lengkap:

$$\text{Similarity}(S, J) =$$

$$\sum (s_k \times j_k) /$$

$$(\sqrt{\sum (s_k^2)} \times \sqrt{\sum (j_k^2)})$$

Skor CBF:

$$\text{Score_cbf} = \text{Similarity}(S, J) + \text{Bonus}$$

3. COLLABORATIVE FILTERING (CF)

Similarity antar siswa:

$$\text{Sim}(a, b) =$$

$$\sum (u_{ak} \times u_{bk}) /$$

$$(\sqrt{\sum (u_{ak}^2)} \times \sqrt{\sum (u_{bk}^2)})$$

Prediksi Skor Jurusan:

$$\text{Score_cf}(i) =$$

$$\sum (\text{Sim}(u, v) \times \text{Score_v}(i)) / \sum \text{Sim}(u, v)$$

4. HYBRID RECOMMENDATION

$$\text{HybridScore}(i) =$$

$$(0.6 \times \text{Score_cbf}(i)) +$$

$$(0.4 \times \text{Score_cf}(i))$$

Jurusan diurutkan berdasarkan HybridScore tertinggi.